Secure User Authentication in Web Applications

Using HTML, CSS, JavaScript, PHP, MySQL & Sessions

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1. Introduction

User authentication verifies the identity of a user before giving access to a web application.
It helps to protect user data and prevents unauthorized access.

2. Problem

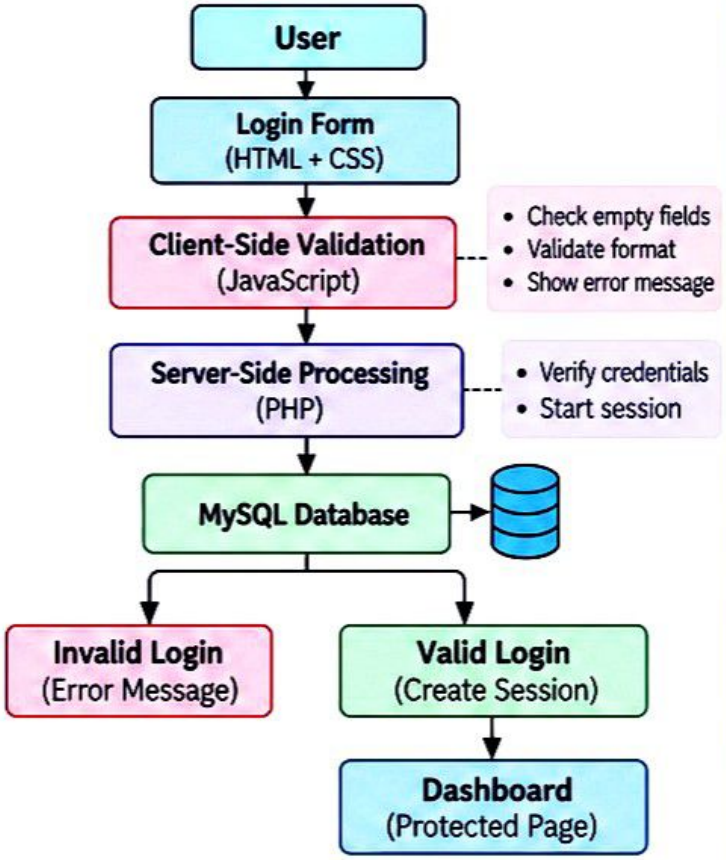
- Unauthorized access to web pages
- Invalid or incorrect login details
- Security threats to user data
- Need to maintain user session



3. Technologies Used

HTML	CSS	JavaScript	PHP	MySQL
Structure of web pages	Design and layout	Client-side validation	Server-side processing	Data storage

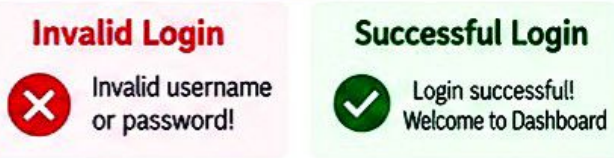
4. System Architecture / Workflow



5. Login Form (UI)



6. Output Screens



7. Database Table (users)

id	username	password	email
1	vaidehi	********	vaidehi@gmail.com
2	meet	********	meet@gmail.com
3	riya	********	riya@gmail.com

8. Code Snippets

HTML (Login Form)

```
<form method="post" action="login.php">
  <input type="text" name="username"
    placeholder="Enter username" required>
  <input type="password" name="password"
    placeholder="Enter password" required>
  <button type="submit">Login</button>
</form>
```

JavaScript (Validation)

```
function validate() {
  let user = document.
      getElementById("username").value;
  let pass = document.
      getElementById("password").value;
  if(user=="" || pass==""){
    alert("Please enter username
     and password!");
     return false;
  }
}
```

PHP (Login & Session)

```
<?php
session_start();
$conn = new mysqli("localhost","root","","students");
$user = $_POST['username'];
$pass = $_POST['password'];
$sql = "SELECT * FROM users
        WHERE username='$user'
        AND password='$pass'";
$result = $conn->query($sql);
if($result->num_rows > 0){
  $_SESSION["user"] = $user;
  header("Location: dashboard.php");
} else {
  echo "Invalid username or password!";
?>
```

9. Security Practices

- Validate input on both client-side and server-side
- Use password hashing (e.g., password_hash())
- Use pessions to maintain login state
- Use prepared statements to prevent SQL injection
- Show proper error messages
- Logout and destroy session when user exits

10. Conclusion

The system provides a simple and secure way to authenticate users using HTML, CSS, JavaScript, PHP, MySQL and sessions. It helps to protect user data and restrict unauthorized access.